Khwezi Mining Health, Safety & Equipment Monitoring Application

Pseudocode

1. **System Overview – Main Program**

```
START
  Generate or load application data
  Initialise user session
  IF user is not logged in THEN
    Display login screen
    INPUT username and password
    IF username and password are correct THEN
      Store username, name and role
      Set login status to TRUE
      Open dashboard
    ELSE
      Display "Invalid username or password"
      Allow user to try again
    ENDIF
  ENDIF
  Display navigation menu for the user's role
  INPUT selected page
  IF user has permission for selected page THEN
    Open selected monitoring page
  ELSE
    Display "Access denied"
  ENDIF
```

IF user selects Logout THEN

Clear stored session information

Return to login screen

ENDIF

END

2. Application Startup and Data Preparation

START

Set application title and page settings

Create "data" folder if it does not exist

Set random seed to 42

SET required files to:
users.json
workers.csv
incidents.csv
equipment.csv
maintenance.csv

FOR EACH required file DO

IF file does not exist THEN

IF file is users.json THEN
Create default user accounts
Hash user passwords
Save users.json

ELSE IF file is workers.csv THEN
Generate 60 sample worker records
Save workers.csv

```
        ELSE IF file is incidents.csv THEN
            Generate 120 sample incident records
            Save incidents.csv

        ELSE IF file is equipment.csv THEN
            Generate 32 equipment records
            Calculate equipment availability
            Save equipment.csv

        ELSE IF file is maintenance.csv THEN
            Generate 40 maintenance records
            Save maintenance.csv
        ENDIF

    ELSE
        Keep existing file unchanged
    ENDIF

ENDFOR

Initialise session variables

Continue to login authentication

END
```

3. <u>User Authentication and Access Control</u>

```
START

    Display login form

    INPUT username
    INPUT password

    Load user accounts from users.json
```

```
    Hash the entered password using SHA-256

    IF username exists AND
      hashed password matches stored password THEN

        Set logged_in to TRUE

        Store username, full name and user role

        Set current page to Dashboard

        Retrieve pages allowed for user role

        Display allowed pages in sidebar

        INPUT selected page

        IF selected page is allowed THEN
            Execute selected page function
            Display page contents
        ELSE
            Display "Access denied"
        ENDIF

    ELSE
        Display "Invalid username or password"
        Return to login form
    ENDIF

    IF Logout button is selected THEN
        Remove username, name, role,
        login status and current page
        Return to login screen
    ENDIF

END
```

4. Worker Safety and Incident Management

A. Worker Safety Monitoring

```
START

  Open Worker Safety page

  Check user permission

  Load workers.csv

  INPUT department filter
  INPUT shift filter
  INPUT risk level filter

  Apply selected filters

  Display filtered worker records

  FOR EACH worker in workers.csv DO

    IF PPE Compliance is "Non-Compliant"
      OR Safety Training is "Expired"
      OR Fatigue Level >= 7 THEN

       Mark worker as unsafe
       Add worker to unsafe workers list

    ENDIF

  ENDFOR

  IF unsafe workers list is not empty THEN
    Display number of unsafe workers
    Display warning message
    Display flagged worker records
  ELSE
    Display "No unsafe conditions detected"
```

```
        ENDIF

    END
```

B. Safety Incident Management

```
START

    Open Safety Incidents page

    Check user permission

    Load incidents.csv

    Display incident management options

    INPUT selected option

    IF option is "View / Filter" THEN

        INPUT department, shift, severity,
            status and search filters

        Filter incident records

        Count matching incidents
        Count major and critical incidents
        Count injuries and lost-time injuries

        Display filtered records and totals

    ELSE IF option is "Add Incident" THEN

        INPUT incident date and time
        INPUT shift and location
        INPUT department and incident type
        INPUT severity and injury details
        INPUT cause and corrective action
```

INPUT incident status

IF Save Incident is selected THEN
Generate new incident ID
Create incident record
Append record to incidents.csv
Display confirmation message
Refresh application data
ENDIF

ELSE IF option is "Trends" THEN

Group incidents by type
Group incidents by month
Group incidents by shift
Group incidents by cause

Display incident trend charts

ENDIF

END

5. Equipment and Maintenance Management

A. Equipment Records

START

Open Equipment page

Check user permission

Load equipment.csv

```
INPUT equipment type filter
INPUT manufacturer filter
INPUT maintenance status filter

Apply selected filters

Display equipment records

IF user selects "Save Equipment" THEN

   INPUT equipment ID
   INPUT equipment type and manufacturer
   INPUT operating hours
   INPUT downtime hours
   INPUT temperature and vibration
   INPUT fuel consumption
   INPUT maintenance status

   TotalHours = OperatingHours + Downtime

   IF TotalHours > 0 THEN
      Availability =
         (OperatingHours / TotalHours) * 100
   ELSE
      Availability = 0
   ENDIF
```

Set brake status to "OK"

Set tyre status to "OK"

Set engine status to "OK"

Create new equipment record

Add record to equipment.csv

Save updated equipment file

Display success message

ENDIF

END

B. Maintenance Monitoring

START

Open Maintenance page

Check user permission

Load equipment.csv

FOR EACH equipment record DO

TotalHours = OperatingHours + Downtime

```
    Availability =
      (OperatingHours / TotalHours) * 100

    Update calculated availability

ENDFOR

INPUT maintenance status filter
INPUT low availability filter

Apply selected filters

Display filtered equipment status

Count equipment due for maintenance
Count overdue equipment
Count equipment under maintenance

FOR EACH equipment record DO

    IF Maintenance Status is "Due"
       OR Maintenance Status is "Overdue"
       OR Downtime > 300
       OR Availability < 80 THEN

        Add equipment to priority list

    ENDIF

ENDFOR

Sort priority list by availability
in ascending order

Display priority maintenance list

Display maintenance counts
```

Display availability chart by equipment type

END

6. **Equipment Condition Monitoring**

START

Open Condition Monitoring page

Check user permission

Load equipment.csv

FOR EACH equipment record DO

READ Temperature T
READ Vibration V

// Temperature classification

IF T > 100 THEN
TemperatureStatus = "CRITICAL"
ELSE IF T >= 80 THEN
TemperatureStatus = "WARNING"
ELSE
TemperatureStatus = "NORMAL"
ENDIF

// Vibration classification

IF V > 8 THEN
VibrationStatus = "CRITICAL"
ELSE IF V >= 5 THEN
VibrationStatus = "WARNING"
ELSE
VibrationStatus = "NORMAL"

```
        ENDIF

        // Overall condition

        IF TemperatureStatus = "CRITICAL"
          OR VibrationStatus = "CRITICAL" THEN

           OverallStatus = "CRITICAL"

        ELSE IF TemperatureStatus = "WARNING"
              OR VibrationStatus = "WARNING" THEN

           OverallStatus = "WARNING"

        ELSE
           OverallStatus = "NORMAL"
        ENDIF

        Store equipment condition status

    ENDFOR

    Count NORMAL equipment
    Count WARNING equipment
    Count CRITICAL equipment

    Display condition monitoring table

    IF critical equipment exists THEN
       Display critical equipment alerts
       Display recommended inspection action
    ELSE
       Display "No critical equipment"
    ENDIF

END
```

7. Alerts and Notification Generation

START

Open Alerts page

Check user permission

Load workers.csv
Load incidents.csv
Load equipment.csv

Create empty alerts list

FOR EACH equipment record DO

Check temperature status

IF temperature is CRITICAL THEN
Add critical temperature alert
ELSE IF temperature is WARNING THEN
Add temperature warning
ENDIF

Check vibration status

IF vibration is CRITICAL THEN
Add critical vibration alert
ELSE IF vibration is WARNING THEN
Add vibration warning
ENDIF

IF Downtime > 300 THEN
Add downtime warning
ENDIF

IF Availability < 80 THEN
Add low availability warning

```
    ENDIF

    IF Maintenance Status is "Overdue" THEN
        Add critical maintenance alert
    ENDIF

ENDFOR

FOR EACH worker record DO

    IF PPE is non-compliant THEN
        Add PPE warning
    ENDIF

    IF Fatigue Level >= 7 THEN
        Add fatigue warning
    ENDIF

    IF Safety Training is expired THEN
        Add training warning
    ENDIF

ENDFOR

FOR EACH incident record DO

    IF Severity is "Critical"
       AND Status is not "Closed" THEN

        Add critical incident alert

    ENDIF

ENDFOR

IF alerts list is empty THEN
    Display "No active alerts"
ELSE
```

```
    Count critical alerts
    Count warning alerts

    Display alert totals

    INPUT alert level and category filters

    Apply selected filters

    FOR EACH matching alert DO
      Display level, category and source
      Display alert message
      Display recommended action
    ENDFOR

  ENDIF

END
```

8. <u>Risk Assessment and Classification</u>

```
START

  Open Risk Assessment page

  Check user permission

  INPUT Likelihood between 1 and 5
  INPUT Consequence between 1 and 5

  RiskScore = Likelihood * Consequence

  IF RiskScore >= 17 THEN
    RiskLevel = "Critical"
    Display "Immediate intervention required"

  ELSE IF RiskScore >= 10 THEN
```

```
    RiskLevel = "High"
    Display "Action required soon"

ELSE IF RiskScore >= 5 THEN
    RiskLevel = "Medium"
    Display "Monitor"

ELSE
    RiskLevel = "Low"
    Display "Acceptable"
ENDIF

Display calculated risk score
Display risk classification

Generate 5 by 5 risk matrix

FOR Likelihood FROM 5 DOWN TO 1 DO
    FOR Consequence FROM 1 TO 5 DO
        MatrixScore = Likelihood * Consequence
        Display MatrixScore
    ENDFOR
ENDFOR

IF user submits Safety Observation form THEN

    INPUT Worker ID
    INPUT observation description
    INPUT observation likelihood
    INPUT observation consequence

    Calculate observation risk score

    Classify observation risk

    Display confirmation message

    IF RiskLevel is "High"
```

OR RiskLevel is "Critical" THEN

Display notification to inform supervisor

ENDIF

ENDIF

END

9. Analytics, Reports and User Administration

A. Data Analytics

START

Open Analytics page

Check user permission

Load workers.csv
Load incidents.csv
Load equipment.csv

Convert incident dates to date format

Display available business questions

INPUT selected question

IF question concerns safety incidents THEN

Apply date filters if required

Count or group incidents by:
Department
Shift
Type
Severity

Calculate requested incident statistics

ELSE IF question concerns worker safety THEN

Calculate PPE compliance percentage

OR calculate total near misses

ELSE IF question concerns equipment THEN

Analyse:
- Downtime
- Availability
- Temperature
- Vibration
- Maintenance status

Calculate requested equipment statistics

ENDIF

Display results using:
- Key performance indicators
- Tables
- Bar charts
- Line charts
- Pie charts

END

B. Report Generation and Data Export

START

Open Reports page

Check user permission

```
  Load workers.csv
  Load incidents.csv
  Load equipment.csv

  IF user selects "Build Report" THEN

    Count total workers
    Calculate PPE compliance percentage
    Count high and critical risk workers

    Count total incidents
    Count major and critical incidents
    Count lost-time injuries

    Count total equipment
    Calculate average equipment availability
    Count overdue maintenance records

    Display report date and time

    Display summary report

  ENDIF

  Display CSV export options

  IF user selects Workers CSV THEN
    Download workers_export.csv
  ELSE IF user selects Incidents CSV THEN
    Download incidents_export.csv
  ELSE IF user selects Equipment CSV THEN
    Download equipment_export.csv
  ENDIF

END
```

C. User Administration

START

Open Manage Users page

Check user permission

Load users.json

Display existing user accounts

INPUT new username
INPUT full name
INPUT password
INPUT user role

IF Create User button is selected THEN

IF username already exists THEN
Display "Username already exists"

ELSE IF username is empty
OR password is empty THEN

Display "Username and password required"

ELSE

Hash password using SHA-256

Create new user account

Assign selected user role

Save updated users.json

Display success message

```
        Refresh application

      ENDIF

    ENDIF

END
```